

VARUN MEDAPPA

CONTACT

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SKILLS

Languages

- Python
- SQL
- Bash
- PyTorch
- TensorFlow

Platforms

- AWS
- Databricks
- GCP

Tooling

- OpenAI APIs (VLM/LLM)
- Embeddings + ANN search
- PEFT / LoRA

ACCOLADES

- 2021 Best AI project award at a Vista-organized hackathon
- Patent for Multimodal Classification:
patents.justia.com/patent/12217493

EDUCATION

Master of Science, Operations Research

Columbia University

Bachelor of Engineering, Industrial Engineering

R. V. College Of Engineering

PROFESSIONAL SUMMARY

Cross-functional Applied Science Leader with 9+ years detecting and investigating harmful content in multimodal audio/video data at scale, working daily in Python and SQL. Founding member and hands-on leader of IAS's Multimodal ML team, whose GARM-aligned detection systems process 3 trillion impressions annually and 50 years of video daily across TikTok, YouTube, Meta, and X, powering ~\$150M in annual revenue (~30% of company). My core work sits at the seam of data science, ML engineering, and policy: translating harm definitions into production classifiers, investigating emerging abuse trends like deepfakes and misinformation, and scaling detection, labeling, and enforcement workflows with LLMs.

EXPERIENCE

Integral Ad Science | New York, NY | 2019 - Present

Staff Applied Scientist - Multimodal Trust & Safety

- Ideated from scratch, built, and shipped ALP, an internal automated labeling platform (text/image/video embeddings + vector index, ANN search, routing positive matches to VLM/LLM review on OpenAI's APIs) that has replaced nearly all manual labeling at IAS, cutting labeling cost and turnaround by 99.5%.
- Scaled ALP into a company-wide capability adopted by multiple teams: managed internal teams as stakeholders, prioritized feature requests and bug fixes against a research roadmap, and drove adoption across the org.
- Served as interim manager of the Innovation team, managing 8 FTE across Trust & Safety Policy, ML research, ML engineering, DevOps, and model deployment, coordinating detection priorities directly with policy, product, and engineering stakeholders.
- Led PEFT/LoRA adoption for A/V deepfake detection, an emerging-abuse workstream, slashing compute ~80% and boosting experiment throughput ~3x without degrading detection performance.

Senior Applied Scientist - Multimodal Trust & Safety

- Founding member of the Innovation/Media team. Developed a multimodal video detection system (TensorFlow, ECS Fargate) processing 50 years of video per day and 3 trillion impressions per year across TikTok, YouTube, Meta, and Twitter, enforcing 12 GARM unsafe-content categories and 37 contextual categories with sustained production monitoring (Patent: US 12,217,493).
- Directed end-to-end development and deployment of a Misinformation Detection initiative, prototyping then productionizing multimodal and DistilBERT models extended for longer token lengths to investigate and flag false and misleading narratives across 22M+ videos per day.

Applied Scientist - Multimodal Trust & Safety

- Developed custom machine translation models (OpenNMT) for 42 language pairs, translating 75 trillion characters annually and cutting costs 67%, enabling multilingual harm detection in production data pipelines.
- Built a pseudo-labeling pipeline that cut model development costs by 90%+ across all unsafe and contextual categories by fine-tuning vision-language models (SigLIP, Vicuna), accelerating detection coverage of new harm verticals.
- Led a team generating synthetic Twitter data with LLMs (GPT-Neo), raising Brand Safety classifier precision by 49% and reducing enforcement false positives.

Associate Applied Scientist - Incrementality/Lift team

- Supported an Online Conversion Lift pipeline ingesting 8 billion impressions per day, applying causal measurement to attribute treatment effects of ad campaigns.

Capillary Technologies | Bangalore | 2016 - 2017

Business Analyst

Mu Sigma | Bangalore | 2015 - 2016

Data Scientist